

INTRODUCTION

Indian life science research is quite significant, serving as a hub for advancements in biotechnology, genomics, agriculture, medicine, and toxicology. DBT, ICMR, and CSIR are the leading institutions promoting life science research. In addition to this, IISERs and several other universities serve as grounds for the development of life sciences. I interviewed several researchers from different institutes, including CSIR-NCL, IISER Pune, IIT Bombay, RIMT University Punjab, and CUSB Gaya. The interviews included general questions about how they feel about their contributions, their methods of communication, and their collaborative approaches in the field of life sciences.

DISCUSSIONS

IIT Bombay

Rutuja Thale

(Research scholar at the Department of Biosciences and Bioengineering)

rutujathale17@gmail.com

1. What are your contributions towards life science (publishing and research) and how do you feel about them?

I did my Master's project on Hydrogel, which has applications in wound healing. I am also a second author on a publication for the same. It feels good that I got this opportunity to pursue this project during my Master's, which gave me an idea about research and also motivated me to continue for a PhD.

2. How do you make others understand your research in the most compelling and understanding manner?

It is very important to convey our research to other people even if they are not from the scientific community. It is our identity, and we should get intimidated by it. In my opinion, we should avoid using technical jargon to explain our research and rather explain in a layman's language so that anyone can understand. Rather than explaining the details of our research, it is better to convey the importance of it.

RIMT University, Punjab

Yuvraj Mehta

yuvrajmehta@rimt.ac.in

(Assistant Professor at the Department of Forensic Science)

1. How do you communicate your research to people?

Last year there was a national seminar about forensic ballistics in which many speakers delivered their lectures on various aspects of forensic ballistics including national and international medicolegal experts.

As a part of life sciences, though we are more inclined towards paramedics, this is our duty to keep the records and information about the latest trends. Being a part of an organization which organizes such events, it feels great.

2. How do you feel about your contributions towards life science?

Indian Life Science Research: An Interview Article

I compared the strengths of dental restoration after exposure to different beverages. Moreover, I worked on the persistence of seminal fluid on different types of clothing materials to deal with rape case investigations. Right now, I am working on the ADMET (Absorption, Distribution, Metabolism and Excretion) prediction of NPS (New Psychotropic Substances). Contributing to forensic science for societal benefit fills me with gratitude, knowing I've served the community meaningfully.

Central University of South Bihar (CUSB), Gaya

Nitin Thakur

nitinthakur@cusb.ac.in

(Research scholar at the Department of Pharmacology)

1. How important are collaborations between academia and industry in advancing life sciences?

The industry needs knowledgeable manpower, and academia needs funds. Both are each other's fulfillers. Together, both can conduct product-oriented research that will benefit both and society.

2. What type of research pique your interest and why?

The effect of biometric nanoparticles on neurological disorders is the field of research that piques my interest. Their small size, biocompatibility, ability to cross barriers like BBB, Blood Brain Barrier, targeted delivery reducing the risk of side effects make them interesting to study.

Central University of Himachal Pradesh

Atin Thakur

Atin.thakur@students.iiserpune.ac.in

(Ex-Intern at Department of Animal Science)

1. Do you have any experience at life science lab?

In December 2024, I worked in the Department of Animal Science at the Central University of Himachal Pradesh (CUHP), gaining hands-on experience in several essential techniques. I learned DNA extraction and used a spectrophotometer for biomolecular analysis. Exploring Pisces toxicology, I studied oxidative stress in zebrafish, which deepened my understanding of its use as a model organism. Its genetic makeup is 70 percent similar to that of humans. It exhibits infinity movements when scared and freezes for some time in cold water, which is fascinating to observe. The long-term impact of toxicants on their bodies highlights the possible prolonged effects on humans, indicating the importance of controlling pollutants in our environment.

2. What is the importance of your understanding of life science with respect to what you have done so far?

As life science researchers, we must work to maintain a sustainable Earth, not only for mankind but also for other organisms. This is where I came across extracting POPs from the soil, which are further sent to higher laboratories for analysis. Additionally, I practiced phytochemical extraction techniques, introducing me to plant-based bioactive compounds. The supportive and collaborative environment at CUHP is exemplified by the fact that there is only one life science laboratory. Despite this, researchers help each other without conflict and focus on the toxicology of organisms and their surroundings, addressing one of the most pressing issues of our time.

CSIR NCL, Pune

(Anonymous fellow)

1. What topics did you share primarily at the conference you attended?

I presented a Poster representation at the PSI conference where I highlighted the importance of protein markers which can detect diabetes; a condition closely linked to cardiovascular diseases. Using mass spectrometry, researchers identified a protein that serves as a marker for high sugar levels, which may contribute to cardiovascular complications. In Type 2 Diabetes Mellitus (T2DM), insulin resistance leads to elevated blood sugar levels, further increasing health risks.

2. How do you communicate your research to people?

The most common way to convey this research in front of people are Poster presentation, Discussion, talking about what the attendees want to know.

Conclusion

The insights from the researcher interviews show the diverse compartments advancing life sciences and, in turn, benefiting society. Their diverse contributions range from hydrogel applications in wound healing to forensic science and diabetes research. Their passion for science communication spans from simplifying concepts for those unfamiliar with the field to advanced discussions for those with a basic understanding. Each one conveys a generalized sense of the need for collaboration and daily improvement.

Our innovations and technologies in life sciences are remarkable and life science is going to boom with the extensive researches going on. Be it limited space or a lack of tools, researchers are dedicated to finding every possible solution. As the field progresses, the collective efforts to understand the research and provide promising solutions to some of the most pressing issues can only be achieved by listening to what researchers need. One such approach I took was to understand how they feel about their research, their methods of communicating it, and their contributions as life science researchers. I hope you like it.

I would like to express my gratitude to all the researchers, especially the individual who remained anonymous, for their valuable work and contributions to the life science.

For any other information or discussions:

atin.thakur@students.iiserpune.ac.in

+91 9015113663

